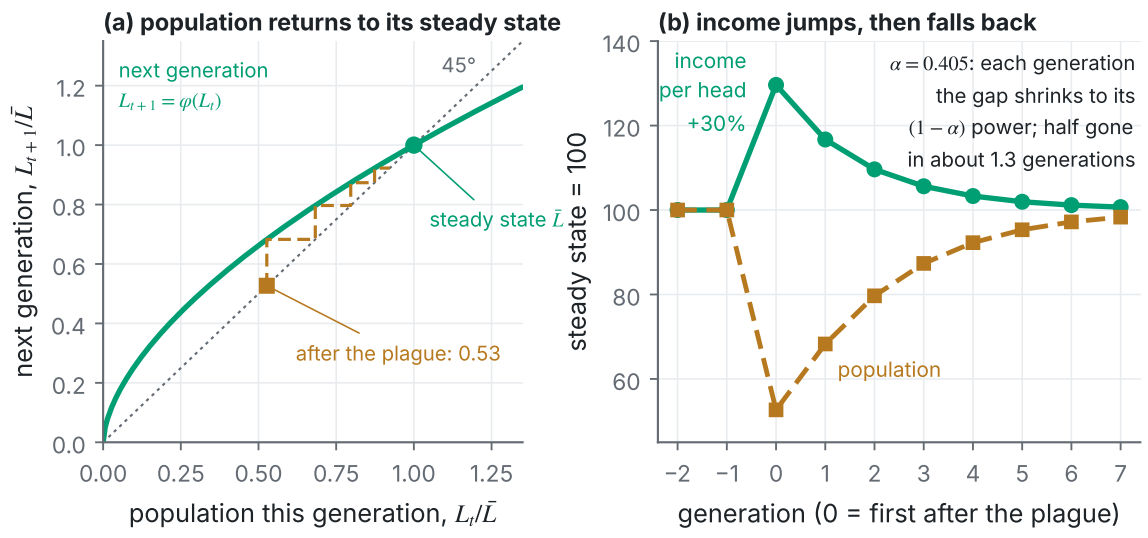
(a) population returns to its steady state
next generation
$L_{t+1} = \varphi(L_t)$
45°
steady state $\bar{L}$
after the plague: 0.53
next generation, $L_{t+1}/\bar{L}$
population this generation, $L_t/\bar{L}$
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.00
0.25
0.50
0.75
1.00
1.25
(b) income jumps, then falls back
income per head +30%
$\alpha = 0.405$: each generation the gap shrinks to its $(1-\alpha)$ power; half gone in about 1.3 generations
population
steady state = 100
60
80
100
120
140
−2
−1
0
1
2
3
4
5
6
7
generation (0 = first after the plague)